

Q Merge two sorted Arr

→ Brute Force T.C $\Rightarrow O(m+n)$
S.C $\Rightarrow O(m+n)$

```
int ans[n];
int i=0, j=0, k=0;
while (i<m || j<n) {
    if (arr1[i] <= arr2[j]) ans[k++] = arr1[i++];
    else ans[k++] = arr2[j++];
}
while (i<m) {
    ans[k++] = arr1[i++];
}
while (j<n) {
    ans[k++] = arr2[j++];
}
```

} S.C is high
we can do it
in $O(1)$

→ Optimize T.C $\Rightarrow O(m+n)$
S.C $\Rightarrow O(1)$

```
i = m-1;
j = n-1;
k = m+n-1;
while (i>=0 || j>=0) {
    if (arr1[i] > arr2[j]) {
        arr[k--] = arr1[i--];
    }
    else {
        arr[k--] = arr2[j--];
    }
}
```

```
while (j>=0) {
    arr[k--] = arr2[j--];
}
```

Dry Run :-

